EN62115 TEST REPORT

Sample Name :	TOY CAR
Trademark :	N/A
Model :	HL1618,HC301,S306,HL2588,JE1005,JJ2164,JJ2199,QY2088,S316, BBH1188,BBH5188,S308,SL400,JT1198,BBH0007,JH101,S307, DK-RRE99, XMX815,628,9188,CCK77336E,XMW911,SH1166,KL1888, SX2038,SX2088, SX2028, KL1666,HL358
Applicant :	SHINE RING TOYS CO.,LTD.
Address :	Room 1603 Building 2 East District Wanlian Business Centre Jiashan Town Zhejiang China
Manufacturer:	SHINE RING TOYS CO.,LTD.
Address:	Room 1603 Building 2 East District Wanlian Business Centre Jiashan Town Zhejiang China
Prepared By :	ShenZhen BYS Testing Co.Ltd
Address :	Room 201, Building A, No. 56, Nanlian Road, Nanlian Community, Longgang Street, Longgang District, Shenzhen
Test Date:	Oct. 25, 2022- Oct. 31, 2022
Date:	Oct. 31, 2022
Report No.:	BYS221025354NR

On Behalf of

TEST REPORT
EN 62115
Electric toys - Safety

Testing Laboratory Name:	Shenzhen BYS Testing Co., Ltd.
Address:	Room 201, Building A, No. 56, Nanlian Road, Nanlian Community, Longgang Street, Longgang District, Shenzhen
Testing location:	Shenzhen BYS Testing Co., Ltd.
Applicant's Name:	SHINE RING TOYS CO.,LTD.
Address:	Room 1603 Building 2 East District Wanlian Business Centre Jiashan Town Zhejiang China
Manufacturer:	SHINE RING TOYS CO.,LTD.
Address:	Room 1603 Building 2 East District Wanlian Business Centre Jiashan Town Zhejiang China
Test specification	
Standard:	EN 62115:2020
Test procedure:	Type test
Procedure deviation:	N/A
Non-standard test method:	N/A
Test Report Form	
Test Report Form No.:	IEC62115_1D
TRF originator:	SEMKO
Master TRF:	dated 20-01

Test item description:	TOY CAR
Trademark:	N/A
Model and/or type reference:	HL1618
	HL1638, BBH1188, BBH5188,BBH0007, SL400, XMX611, JT1198, JT2199, JJ2168, XMX815, 628, 9188, 9166, 1919A, 9936, 938, 911, 2188, 2988, 688, 201, 7188, 968, 988,911, 808, 988, 989
Rating(s):	AC 220-240, 50/60Hz,

Test case verdicts

Test case does not apply to the test object .. :	N/A
Test item does meet the requirement .... :	P(ass)
Test item does not meet the requirement ..... :	F(ail)

Tested By (Test Engineer)	:	
Reviewed By (Supervisor)	:	
Approved By (Chief Engineer)	:	

Shenzhen BYS Testing Co. LTD.
BYS
APPROVED

General remarks

"(See Enclosure #)" refers to additional information appended to the report.

"(See appended table)" refers to a table appended to the report.

Throughout this report a ☐ comma / ☒ point is used as the decimal separator.

General product information:

1. This TOY CAR head unit is supplied by lead storage battery.
2. All components are mounted on the PCB and housed with plastic enclosure.
3. The maximum operating temperature of the product is 45°C.
4. The unit weight: Below 2kg.

All models are identical except for the name and appearance. All tests are performed on WJ016.

Artwork of marking plate:

TOY CAR

Model: HL1618

AC 220-240, 50/60Hz,

CE

SHINE RING TOYS CO.,LTD.

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Cl.	Requirement – Test	Result	Verdict

5	**GENERAL CONDITIONS FOR THE TESTS**		—
5.1	General		P
	The tests are carried out in a draught-free location at an ambient temperature of 20 °C ±5 °C.		P
	Electric toys designed to bear the mass of a child are loaded with – 25 kg, if intended for children up to 3 years old; – 50 kg, if intended for older children.		P
5.2	Preconditioning		P
	The preconditioned sample is prepared by subjecting it to the following test methods of ISO 8124- 1 in the order specified below, with batteries in position:		P
	– Tension test – for all Electric toys; however, the force being 70 N ±2 N independent of the dimensions and applicable independent of age group;		P
	– Drop test – from 93 cm ±5 cm, irrespective of the age group. The drop test is not carried out on large and bulky Electric toys;		N/A
	– Tip over test – for large and bulky Electric toys;		P
	– Static strength test – for Electric toys designed to bear the mass of a child;		P
	– Dynamic strength test – for wheeled ride-on Electric toys;		P
	– Tension test for seams and materials – for Electric toys having textile or other flexible materials covering batteries or other electricalparts.		N/A
5.3	Assembly		P
	If an electric toy is intended to be assembled by a child, the requirements apply to each part accessible to the child and to the assembled electric toy. If an electric toy is intended to be assembled by an adult, the requirements apply to the assembled electric toy.		P
5.4	Movable parts		P
	Tests are carried out with the electric toy or any movable part of it placed in the most unfavourable position when the electric toy is played with as intended or in any foreseeable way.		P
5.5	Detachable parts		P
	Detachable parts shall be removed or left in place whichever gives the most unfavourable conditions.		P
5.6	Settings		N/A

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Cl.	Requirement – Test	Result	Verdict
	Electric toys provided with controls or switching devices are tested with these controls or devices adjusted to their most unfavourable setting, if the setting can be altered by theuser.		N/A
5.7	Selection of power supplies		P
5.7.1	General		P
	Electric toys having one or more type of supply shall be tested using each type of supply or any combination of those supplies, whichever creates the most unfavourable condition.		P
	Protection in the power supply shall be disabled unless the power supply is specifically designed for the electric toy and cannot be replaced by other power supplies.		P
	Electric toys having more than one rated voltage or a range of rated voltages are tested at the most unfavourable voltage.		N/A
	Electric toys using an AC input only are tested with AC at rated frequency. Those using AC and DC are tested at the most unfavourable frequency or supply condition. If no rated frequency of the supply is marked on the electric toy, the electric toy is tested with 50 Hz or 60 Hz, whichever is more unfavourable.		N/A
	For Electric toys using transformers, power supplies or battery chargers, the tests shall be carried out with the transformer, power supply or battery charger both connected and disconnected.		P
5.7.2	Electric toys that are used with batteries		P
	Unless otherwise specified in the tests, Electric toys that are used with batteries are tested using new alkaline primary batteries. If the manufacturer specifies a different battery technology, chemistry or type the tests shall be repeated with the specified battery in addition to the battery required for the test.		P
	Primary batteries used are those with the voltage and size specified on the toy or in the instructions and shall comply with the relevant parts of the IEC 60086 series IEC. Secondary batteries used for testing shall comply with IEC 62133. Tests are carried out with secondary batteries fully charged.	lead acid battery provided.	N/A
	The tests shall also be carried out with one or more batteries reversed unless this is prevented by the construction, physically or by preventing an electrical connection.		P
5.7.3	Electric toys using battery boxes		P

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	Electric toys intended for use with a battery box are tested with the battery box supplied with the toy or with the battery box recommended in the instructions.		N/A
5.7.4	Electric toys using transformers and power supplies		N/A
	Electric toys using a transformer or a power supply are tested with the transformer or power supply supplied with the electric toy.		N/A
	If the electric toy is supplied without a transformer or power supply, it is tested with the transformer or power supply recommended in theinstructions.		N/A
5.7.5	Electric toys using rechargeable batteries		P
	Electric toys that are used with rechargeable batteries that can be operated during charging shall be tested according to 5.7.4 or with the batteries fully charged, whichever is more unfavourable for each test.		P
5.7.6	Electric toys using other power supplies		N/A
	Electric toys using a USB connection for a power supply shall be supplied at 5 V.		N/A
5.8	Accessories and parts		N/A
	When accessories are made available by the manufacturer, the electric toy is tested with those accessories that give the most unfavourable results. Such parts do not need to have an electrical function to be considered for the testing.		N/A
	Lamps that can be accessed and removed without the aid of a tool are tested with lamps of the highest wattage that can be fitted, irrespective of any marking.		N/A
6	**CRITERIA FOR REDUCED TESTING**		—
6.1	General For some Electric toys, it is not necessary to carry out all the tests specified in this standard if the conditions of 6.2, 6.3 or 6.4 are met.		P
6.2	Short-circuit resistance		P
	For Electric toys that comply with the tests of Clause 9, except 9.6 and 9.8, with any insulation between parts of different polarity short-circuited in turn 13.3, 13.7 and Clauses 8, 9 except 9.6 and 9.8, 10, 11, 17 and 18 are not applicable. The short circuit may be applied by a flexible wire or any other suitable means.		P
6.3	Low power Electric toys		N/A

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Cl.	Requirement – Test	Result	Verdict
	Electric toys that meet both of the following criteria are considered to comply with the requirements of Clauses 9, 10 and 18 if they		N/A
	are supplied by a power source supplying less than 15 W; have an over temperature or over current protection where the clearance between conductive parts of different polarity is at least 3,8 mm for any part of the circuit between the power source and the protection.		N/A
	When a short-circuit is applied to any part of the circuit after the protection, the electric toy shall comply with 9.10.		N/A
6.4	Battery circuits		N/A
	Circuits where the only power source comprises three batteries or less that are of the following designations: – R44/LR44 – R41/LR41 – LR1130 – LR54 are considered to comply with the requirements of Clause 9, 10, 11, 17 and 18.		N/A

7	**MARKING AND INSTRUCTIONS**		P
7.1	General		P
	Instructions and other text required by this standard shall be written in an official language of the country in which the appliance is to besold.	English instructions provided	P
7.2	Markings on Electric toys		P
7.2.1	Electric toys or their packaging are marked with:		P
	– the name, trade mark or identification mark of the manufacturer or responsible vendor......... :	see copy of marking plate.	P
	– the model or type reference .... :	see copy of marking plate.	P
	When the toy is marked, these markings are on the main part.		P
	When the packaging is not marked and when it is not practical to mark the toy, e.g. due to its size, the markings may be contained in the instructions instead.		P
7.2.2	Battery Electric toys with replaceable batteries are marked with:		P
	– the nominal battery voltage, in or on the battery compartment (V)		P
	– the symbol for d.c., if the toy has a battery box..		N/A

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	The electric toy shall be marked with the shape of the batteries, together with the nominal voltage and polarity. The positive terminal shall be indicated by symbol IEC 60417-5005 (2002-10).		P
7.2.3	Electric toys supplied by a transformer or a power supply are marked with:		N/A
	– their rated voltage, (V)....................................:		--
	– the symbol for a.c. or d.c., as applicable; ....:		--
	– their rated power input, if greater than 25 W or 25 VA in		N/A
	– watts (W) ....:		--
	– volt-amperes (VA) ....:		--
	– the symbol for transformer for Electric toys.		N/A
	This symbol is also marked on the packaging		N/A
	The marking of rated voltage and the symbol for a.c. or d.c. is placed adjacent to the terminals.		N/A
	The marking for a.c. or d.c. is not required if the incorrect supply does not impair compliance with this standard.		N/A
	Electric toys intended to be supplied from a power supply for the purposes of recharging the battery shall be marked with symbol IEC 60417-6181 (2016-01) and its type reference along with symbol ISO 7000-0790 or with the substance of the following: "Use only with <model designation> <supply>"		N/A
7.2.4	Dual-supply Electric toys are be marked with the marking required for both battery Electric toys and transformer Electric toys.		N/A
7.2.5	The identification for detachable lamps is be marked with		N/A
	– the rated voltage (V) ....:		--
	– type number ....:		--
	– the maximum power input (W)...................:		--
	– the maximum current. (A)...........................:		--
	The marking for power input or current of detachable lamps is as follows:		N/A
	lamp max (W) ....:		--
	lamp max (A) ....:		--
	The word "lamp" may be replaced by symbol 5012 of IEC 60417-1.		N/A
	The marking is visible when replacing the lamp.		N/A

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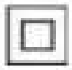
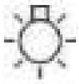
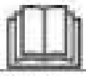
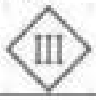
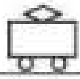
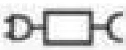
	This marking is not required if the temperature rises measured during the tests of Clause 9 do not exceed the limits when a lamp having the highest power input is fitted.		N/A
7.2.6	When symbols are used, they are as follows:		P
	Symbol 5031 of IEC 60417.		P
	Symbol 5032 of IEC 60417. ~		N/A
	Symbol 5172 of IEC 60417. 		N/A
	Symbol 5012 of IEC 60417. 		N/A
	Symbol 0790 of ISO 7000 		N/A
	Symbol 5180 of IEC 60417 		N/A
	Symbol 5219 of IEC 60417. 		N/A
	Symbol 5005 of IEC 60417. +		P
	Symbol 5006 of IEC 60417. —		P
	Symbol 6181 of IEC 60417. 		N/A
	Symbol ISO 7010 W001. 		N/A

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Cl.	Requirement – Test	Result	Verdict
	Units of physical quantities and their symbols are those of the international standardized system.		P
7.2.7	Durability		P
	The markings on an electric toy shall be legible and durable.		P
7.3	Instructions and markings on packaging		P
7.3.1	Instructions are provided that give details concerning cleaning and maintenance when necessary for the safe operation of the toy.		P
	Electric toys are provided with instructions for assembly if		P
	– they are intended to be assembled by a child;		P
	– these instructions are necessary for safe operation of the toy.		P
	If the toy is intended to be assembled by an adult, this is stated.		P
	The instructions may be on a leaflet, on the packaging or on the electric toy. If the instructions are marked on the electric toy, they shall be visible from the outside and if the electric toy consists of more than one part, only the main part needs to be marked.		P
	Instructions for Electric toys intended to be used in water shall state that the electric toy is to be operated in water only when fully assembled in accordance with the instructions, if applicable.		N/A
	If markings or instructions stated in 7.2 are on the packaging only, they shall be accompanied by a statement indicating that packaging must be retained since it contains important information. If markings or instructions stated in 7.2 are on instruction sheet only, they shall be accompanied by a statement indicating that instruction sheet must be retained since it contains important information.		P
	If part of markings or instructions stated in 7.2 are on packaging and others are in instruction sheet, a statement indicates that instruction sheet and packaging must be retained since it contains important information.		P
7.3.2	Transformer Electric toys and power supply Electric toys		N/A

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Cl.	Requirement – Test	Result	Verdict
	The instructions for Electric toys using a transformer or a power supply or a battery charger shall state that the transformer, power supply or battery charger used with the electric toy shall be regularly examined for damage to the supply cord, plug, enclosure or other parts, and in the event of damage, it shall not be used until the damage has been repaired.		N/A
	For Electric toys using a transformer or a power supply, the following age warning shall be visible to consumers at the time of purchase:		N/A
	WARNING: Not suitable for children under 3 years		N/A
	The text "Not suitable for children under 3 years" may be replaced by the age warning symbol from ISO 8124-1.		N/A
	This requirement does not apply to Electric toys which, on account of their function, dimensions, properties and similar characteristics, are clearly unsuitable for children under 3 years.		N/A
	The instructions for Electric toys using a transformers or a power supply shall state that the toy is not to be connected to more than the recommended number of transformers or power supplies		N/A
	The instructions for Electric toys using a transformers or a power supply shall contain the substance of the following, as applicable:		N/A
	– the toy shall only be used with a transformer for Electric toys or a power supply for Electric toys (as applicable);		N/A
	– the toy shall be used with the transformer or power supply supplied, if the transformer is supplied with the toy;		N/A
	– the model number or specification of a suitable transformer or power supply for use with the toy, if not supplied with the toy;		N/A
	– the transformer or power supply (as applicable) is not a toy;		N/A
	– Electric toys liable to be cleaned with liquids are to be disconnected from the transformer or power supply before cleaning.		N/A
7.3.3	Electric toys that are used with replaceable batteries		P
7.3.3.1	The instructions for Electric toys that are used with replaceable batteries shall contain the substance of the following, as applicable:		P
	– how to remove and insert the batteries;	Described in the instructions	P

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Cl.	Requirement – Test	Result	Verdict
	– non-rechargeable batteries are not to be recharged;	Described in the instructions	P
	– for Electric toys using rechargeable batteries, the batteries should be charged under adult supervision. For batteries charged using a battery charger for use by children, this instruction may be replaced by: "Batteries are only to be charged by persons of at least 8 years old";		N/A
	– different types of batteries or new and used batteries are not to be mixed;		P
	– batteries are to be inserted with the correct polarity (+ and –);		P
	– exhausted batteries are to be removed from the toy;		P
	– the supply terminals are not to be short-circuited.		N/A
	The instructions for Electric toys supplied by a battery box shall state that the toy is not to be connected to more than the recommended number of power supplies.		N/A
	The instruction need not be added if the connections cannot be made easily without the aid of a tool and using parts from two identical Electric toys or constructional sets.		N/A
	The instructions for Electric toys containing non- replaceable batteries shall state the substance of the following: This toy contains batteries that are non-replaceable.		N/A
	The instructions for Electric toys intending to be supplied from a detachable power supply for the purposes of recharging the battery, the type reference of the detachable power supply shall be stated along with the substance of the following: WARNING: For the purposes of recharging the battery, only use the detachable supply unit provided with this toy.		N/A
7.3.3.2	Coin batteries		N/A
	Packaging: Electric toys using replaceable coin batteries shall carry the substance of the following warning on the packaging:		N/A
	WARNING: Contains coin battery. Hazardous if swallowed – see instructions.		N/A

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	Alternatively, the packaging shall be marked with symbol ISO 7000-0790 and warning sign ISO 7010 W001 in conjunction with a supplementary sign containing a coin battery symbol. The combination sign is to meet the rules in ISO 3864-1. This sign shall be placed next to symbol ISO 7000-0790. The meaning of this symbol combination shall be explained in the instructions.		N/A
	Instructions: Electric toys using replaceable coin batteries shall carry the substance of the following warning in the instructions:		N/A
	WARNING: This product contains a coin battery. A coin battery can causeserious internal chemical burns if swallowed.		N/A
	WARNING: Dispose of used batteries immediately. Keep new and used batteries away from children. If you think batteries might have been swallowed or placed inside any part of the body, seek immediate medical attention.		N/A
7.3.3.3	Button batteries		N/A
	Electric toys using replaceable button batteries shall carry the substance of the following warning in the instructions:		N/A
	WARNING: Dispose of used batteries immediately. Keep new and used batteries away from children. If you think batteries might have been swallowed or placed inside any part of the body, seek immediate medical attention.		N/A
7.4	Instructions for Electric toys that can be connected to class I equipment		N/A
	For Electric toys that can be connected to class I equipment which do not meet the requirement of 13.9, the instructions shall state the substance of the following:		N/A
	This toy is only to be connected to equipment bearing either of the followingsymbols: 		N/A
7.5	Instructions for ride-on Electric toys		N/A
	The instructions for ride-on Electric toys, shall carry the substance of the following warning:		N/A
	WARNING: Not to be used in traffic		N/A

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Cl.	Requirement – Test	Result	Verdict
	In addition, the instructions for ride-on Electric toys, shall carry the substance of a warning, preceded by the word "WARNING", which draws attention to the potential hazards of using the ride-on Electric toys in areas other than private grounds.		N/A
7.6	Temperature warnings		N/A
	The accessible parts of Electric toys that are intended for children 3 years and over but less than 8 years which exceed the temperature rise limit for children less than 3 years according to Table 1 (see 9.10) shall carry the following warning that shall be visible to consumers at the time of purchase:		N/A
	WARNING: Not suitable for children under 3 years		N/A
	The text "Not suitable for children under 3 years" may be replaced by the age warning symbol from Figure B.1 of ISO 8124-1.		N/A
	The accessible parts of Electric toys that are intended for children 8 years and over, and which exceed the temperature rise limit for children 3 years to less than 8 years according to Table 1 (see 9.10) shall carry the following warning that shall be visible to consumers at the time of purchase:		N/A
	WARNING: Not suitable for children under 8 years		N/A

8	**POWER INPUT**		—
	The power input of transformer Electric toys and dual supply Electric toys do not exceed the rated power input by more than 20 %.		N/A
	Compliance is checked by measurement when the power input has stabilized and the toy has attained normal operating temperature with		N/A
	– all circuits that can operate simultaneously being in operation;		N/A
	– the toy being supplied at rated voltage;		N/A
	– the toy being operated under normal operation.		N/A
9	**HEATING AND ABNORMAL OPERATION**		P
9.1	Electric toys do not attain excessive temperatures in use.		P

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	They are constructed so that the risk of fire, mechanical damage impairing safety or other hazards, as a result of careless use or failure of a component, is obviated as far as is practicable.		P
	All Electric toys are subjected to the tests of 9.3 to 9.5.		P
	Electric toys incorporating motors are subjected to the test of 9.6.		N/A
	Transformer Electric toys, dual supply Electric toys and Electric toys with battery boxes are subjected to the test of 9.7.		N/A
	Electric toys supplied with power from a USB connection are also subjected to the tests of 9.8.		N/A
	Electric toys incorporating electronic circuits are subjected to the test of 9.9.		N/A
	All Electric toys shall be tested under the conditions specified in 9.2.		P
	If a heating element or an intentionally weak part becomes permanently open-circuited, the relevant test is repeated on a second sample. This second test shall be terminated in the same mode unless the test is otherwise satisfactorily completed. Subsequent tests for Electric toys where a heating element or an intentionally weak part becomes permanently open-circuited shall be completed on a new sample.		N/A
	Damage caused by a short-circuit that does not impair compliance with this standard is repaired before a further short-circuit is applied.		P
	If during the tests of 9.9 an electronic circuit prevents the hazardous conditions listed in 9.10 or dangerous malfunction, it shall additionally comply with Annex D. In this case, the electronic circuit is considered as a protective electronic circuit.		N/A
	Electric toys with an electronic off-mode or stand- by mode shall also comply with Annex D, if the electric toy can malfunction in such a way as to cause any unintended operation that may impair safety.		N/A
	If more than one of the tests are applicable to the same electric toy, these tests are made consecutively after the electric toy has cooled down to room temperature.		P
	Unless otherwise specified, after the tests of 9.3 to 9.9 the electric toy shall comply with 9.10.		P

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Cl.	Requirement – Test	Result	Verdict
9.2	Testing condition		P
9.2.1	Electric toys are placed in the most unfavourable position that can occur during play.		P
	Hand-held Electric toys are freely suspended.		N/A
	Other Electric toys are placed on the floor of a test corner as near to the walls as possible or away from the walls, whichever is more unfavourable.		P
	Electric toys having dimensions not exceeding 500 mm are completely covered with the cotton gauze.		N/A
9.2.2	Battery Electric toys are supplied at rated voltage.		P
	Transformer Electric toys and dual supply Electric toys are supplied at 0,94 times or 1,06 times rated voltage, whichever is more unfavourable.		N/A
9.2.3	The temperature rises are determined by means of fine-wire thermocouples positioned so that they have minimum effect on the temperature of the part under test.		P
	Where thermocouples cannot successfully measure the maximum temperature during the test, thermal paper or other methods to measure temperature rise may be used.		N/A
	Used measurement method.............................:	Thermocouples	N/A
9.2.4	Mobile Electric toys are tested in whichever use condition will create the highest temperature rise.		P
	When non-self-resetting thermal cut-outs operate, they are re-set a maximum of three times.		P
	Electric toys with self-resetting thermal cut-outs are tested until steady state conditions are established.		N/A
9.3	Electric toys are operated under normal operation and the temperature rises of the various parts are determined.		P
	Rechargeable battery Electric toys that can operate during recharging are also tested in the charging mode.		N/A
9.4	Normal operation with insulation short-circuited		N/A
9.4.1	The test of 9.3 is repeated, the insulation between parts of different polarity, except those in battery compartments, being short circuited in turn if it is accessible after the removal of detachable parts, except lamps.		N/A

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Cl.	Requirement – Test	Result	Verdict
	The short circuit is not applied to:		N/A
	lamps and lamp holders, battery compartments, complying with 13.4, other parts where access is only possible by removing covers that can only be removed with the aid of a tool or by two independent movements applied simultaneously.		N/A
9.4.2	Steel pin test		N/A
9.4.3	Steel rod test		N/A
9.5	Abnormal operation with temperature controls made inoperable		N/A
	The test of 9.3 is repeated, any control that limits the temperature during the tests of 9.3 and 9.4 being short-circuited.		N/A
	If the toy has more than one control, they are short- circuited in turn.		N/A
	If the control consists only of positive temperature co-efficient resistors (PTCs), negative temperature co-efficient resistors (NTCs) or voltage dependent resistors (VDRs) they are not short-circuited if they are used within their manufacturers declared specification.		N/A
9.6	Electric toys with accessible moving parts locked		N/A
	The test of 9.3 is repeated with accessible moving parts locked.		N/A
9.7	Additional transformers and power supplies		N/A
9.7	Transformer Electric toys, dual supply Electric toys and Electric toys with battery boxes are connected to a power supply in addition to that recommended in the instructions for use.		N/A
	The additional power supply is identical to that recommended for the toy and is connected in series or in parallel, whichever is more unfavourable.		N/A
	The toy is then tested as specified in 9.3 and 9.4.		N/A
9.8	Abnormal supply to Electric toys via a USB connection		N/A
	For Electric toys supplied with power from a USB connection, the test of 9.3 is repeated with the toy being supplied with a voltage of 42 V.		N/A
9.9	Fault condition in electronic circuits		P
	Fault conditions a) to f) are not applied to circuits or parts of circuits where both of the following conditions are met:		P
	– the electronic circuit is a low-power circuit as described below;		P

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	– the protection against fire hazard or dangerous malfunction in other parts of the toy does not rely on the correct functioning of the electronic circuit.		P
	The following fault conditions are considered and, if necessary, applied one at a time, consequential faults being taken into consideration:		N/A
	a) short circuit of clearances and creepage distances between parts of different polarity, if these distances are less than the values specified in Clause 18, unless the relevant part is adequately encapsulated;		N/A
	b) open circuit at the terminals of any component;		N/A
	c) short circuit of capacitors, unless they comply with IEC 60384-14 or they are ceramic capacitors used within the manufacturer's specification;		N/A
	d) short circuit of any two terminals of an electronic component, other than integrated circuits;		N/A
	e) failure of triacs in the diode mode;		N/A
	f) failure of an integrated circuit. In this case the possible hazardous situations of the toy are assessed to ensure that safety does not rely on the correct functioning of such a component. All possible output signals are considered under fault conditions within the integrated circuit. If it can be shown that a particular output signal is unlikely to occur, then the relevant fault is not considered.		N/A
	In addition, each low-power circuit is short-circuited by connecting the low-power point to the pole of the supply from which the measurements were made.		N/A
	For simulation of the fault conditions, the toy is operated under the conditions specified in 9.2 but supplied at rated voltage..		N/A
	For products that have to be kept switched on by hand or foot, if the applied fault-condition results in the product not functioning, the switch is released after 30 s.		
	If the toy incorporates an electronic circuit that operates to ensure compliance with 9.5 to 9.7, the relevant test is repeated with a single fault simulated, as indicated in a) to f) above.		N/A
	Fault condition f) is applied to encapsulated and similar components if the circuit cannot be assessed by other methods.		N/A

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	PTC resistors are not short-circuited if they are used within the manufacturer's specification.		N/A
	PTC-S thermistors are short-circuited unless they comply with IEC 60738-1.		N/A
9.10	Compliance criteria		P
	During the tests, the temperature rises of accessible parts are monitored continuously.		P
	The temperature rise of the surface of handles, knobs and other parts that are likely to be touched by hand do not exceed the following values:		P
	However, during the test of 9.8, the temperature rise of accessible parts of the electric toy shall not exceed 1,5 times the values specified in Table 1.		N/A
	The temperature rise of battery surfaces and other parts inside the battery compartment, where batteries are inside a battery compartment with a cover, which can only be opened by the use of a tool or by at least two independent movements applied simultaneously, shall not exceed 45 K.		P
	During the tests:		P
	– sealing compound shall not flow out;		P
	– vapour shall not accumulate in the electric toy;		P
	– dangerous substances shall not be produced, such as poisonous or ignitable gas, in hazardous amounts;		P
	– enclosures shall not deform to such an extent that compliance with this International Standard is impaired;		P
	– batteries shall not leak fluids or erupt;		P
	– materials, including the cotton gauze, shall not char;		P
	– the electric toy shall not emit flames or molten metal.		P
	After the tests, the electric toy shall not be damaged to such an extent that compliance with the standard is impaired.		P
	Electric toys having accessible parts with temperature rises exceeding the values in Table 1 for children less than 3 years or for children between 3 years and 8 years shall have a warning together with the appropriate age indication, 3 years or 8 years (see 7.6).		N/A
10	**ELECTRIC STRENGTH**		N/A

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Cl.	Requirement – Test	Result	Verdict
10.1	Electric strength at operating temperature		N/A
	The electrical insulation of the toy at operating temperature was adequate.		N/A
	The electric toy is brought to operating temperature and the insulation is immediately subjected to a voltage having a frequency of 50 Hz or 60 Hz for 1 min, in accordance with IEC 61180. The test voltage is 250 V.		N/A
	The test voltage is applied between power input terminals, such as terminals in the battery compartment or power supply input connector terminals and accessible parts, non-metallic accessible parts being covered with metal foil.		N/A
	Electric toys that are used with batteries are tested with the batteries removed.		N/A
	No breakdown occurred.		N/A
10.2	Electric strength under humid conditions		N/A
	The electric insulation of the electric toy under humid conditions shall be adequate.		N/A
	Detachable parts are removed and subjected, if necessary, to the humidity test with the main part.		N/A
	The humidity test is carried out for 48 h in a humidity cabinet containing air with a relative humidity of (93 ±3) %. The temperature of the air is maintained within 2 K of any convenient value t between 20 °C and 30 °C.		N/A
	The electric toy is immediately subjected to a voltage having a frequency of 50 Hz or 60 Hz for 1 min, in accordance with IEC 61180. The test voltage is 250 V.		N/A
	The test voltage is applied between power input terminals, such as terminals in the battery compartment or power supply input connector terminals and accessible parts, non-metallic accessible parts being covered with metal foil.		N/A
	Electric toys that are used with batteries are tested with the batteries removed.		N/A
	No breakdown occurred.		N/A
11	**ELECTRIC TOYS USED IN WATER, ELECTRIC TOYS USED WITH LIQUID AND ELECTRIC TOYS CLEANED WITH LIQUID**		N/A

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Cl.	Requirement – Test	Result	Verdict
	Battery Electric toys intended to be used in water and Electric toys likely to be cleaned with liquid have an enclosure providing the appropriate protection.		N/A
	Compliance for Electric toys intended to be used with liquid and Electric toys intended to filled from a tap is checked by the following test. The electric toy is placed in the filling position according to the instructions and detachable parts are removed. The liquid container of the electric toy is completely filled with water containing approximately 1 % NaCl and a further quantity, equal to 15 % of the capacity of the container or 0,25 l, whichever is the greater, is poured in steadily over a period of 1min.		N/A
	The appliance shall then withstand the electric strength test of 10.1 and inspection shall show that there is no trace of water on insulation that could result in a reduction of clearances or creepage distances below the values specified in Clause 17.		N/A
	Compliance for Electric toys intended to be cleaned with liquid is checked by the test of 14.2.4 of IEC 60529:1989, detachable parts having been removed. The battery cover and other covers shall not be removed when those covers are designed for protection from water.		N/A
	Excess water is then removed from the enclosure. The electric toy shall withstand the electric strength test of 10.1 and inspection shall show that there is no trace of water on insulation that could result in a reduction of creepage distances and clearances below the values specified in Clause 17.		N/A
	Compliance for Electric toys intended to be used in water is checked by the following test, detachable parts requiring a tool to remove them being left in place.		N/A
	The electric toy is immersed in water containing approximately 1 % NaCl, all parts of the electric toy being at least 150 mm below the surface. The electric toy is positioned in the most unfavourable orientation and operated for 15 min ±1 min. There shall be no overpressure within the enclosure due to entrapped gas.		N/A

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Cl.	Requirement – Test	Result	Verdict
	The electric toy is then taken out of the water, positioned to allow excess water to drain, and the enclosure is wiped dry. The electric toy shall withstand the electric strength test of 10.1.		N/A
12	**Mechanical strength**		P
	Enclosures have adequate mechanical strength.		P
	Compliance is checked by applying blows to the appliance in accordance with test Ehb of IEC 60068-2-75, the spring hammer test.		P
	The electric toy is rigidly supported and three blows, having an impact energy of 0,5 J, are applied to every point of the enclosure that is likely to be weak.		P
	Electric toys that are used with batteries are tested with the batteries in place. The blows are not applied to the batteries.		P
	If necessary, the blows are also applied to handles, levers, knobs and similar parts and to signal lamps and their covers but only if the lamps or covers protrude from the enclosure by more than 10 mm or if their surface area exceeds 4 cm2. Lamps within the appliance and their covers are only tested if they are likely to be damaged in normal use.		N/A
	After the test, the electric toy shall show no damage that could impair compliance with 9.3, 9.5, 9.7, 9.8, 13.4.1, 13.4.2, 13.4.3 and 13.6 and Clauses 10, 11, 14 and 17 shall not be impaired.		P
	Damage to the finish, small dents that do not reduce clearances or creepage distances below the values specified in Clause 17, and small chips that do not adversely affect protection against access to live parts or moisture, are ignored.		P
	If a decorative cover is protected by an inner cover, fracture of the decorative cover is ignored if the inner cover itself withstands the test.		N/A
	If there is doubt as to whether a defect has occurred by the application of the preceding blows or the previous tests, this defect is neglected and the group of three blows is applied to the same place on a new sample which shall then withstand the test.		N/A
	After testing, cracks not visible to the naked eye and surface cracks in fibre-reinforced mouldings and similar materials are ignored.		P

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Cl.	Requirement – Test	Result	Verdict
12.2	Attachment strength		N/A
	Non-detachable parts that prevent contact with moving parts or hot surfaces, or access to locations where explosion or fire could be initiated, shall be fixed in a reliable manner and shall withstand the mechanical stress occurring during normal use.		N/A
	Compliance is checked by applying the following pull force:		N/A
	– 50 N, if the longest accessible dimension of the part does not exceed 6 mm;		N/A
	– 90 N, for other parts.		N/A
	The force is gradually applied during a period of 5 s and maintained for a further 10 s.		N/A
	The part shall not become detached.		N/A
13	**CONSTRUCTION**		P
13.1	Nominal supply voltage		P
	Their supply voltage does not exceed 24 V.	Supplied by two AAA batteries	P
	Supply voltage (V)..:	1.5Vx2	-
	The working voltage between any two accessible parts of the toy does not exceed 24 V when the toy is supplied at rated voltage.		P
	Working voltage (V)...:	Less than 24V	--
13.2	Transformers, power supplies and battery chargers		N/A
13.2.1	Mains connections		N/A
	Battery chargers, transformers, power supplies and other parts connected to mains voltage supply shall not be an integral part of the electric toy.		N/A
	Controls for the electric toy shall not be incorporated in the transformer or power supply.		N/A
13.2.2	Electric toys for use in water or for use with liquids		N/A
	Electric toys for use in water and Electric toys for use with liquid shall not require a connection to a transformer, power supply or battery charger in order to work in the water or with the liquid.		N/A
13.2.3	Electric toys for children under the age of 3 years		N/A
	Electric toys using transformers and power supplies shall not be intended for use by children under 3 years.		N/A

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Cl.	Requirement – Test	Result	Verdict
13.3	Thermal cut-outs		N/A
	Non-self-resetting thermal cut-outs, necessary for compliance with this standard, shall only be resettable with the aid of a tool.		N/A
13.4	Batteries		P
13.4.1	Small batteries		N/A
	Batteries that fit wholly within the small parts cylinder as specified in 5.2 of ISO 8124-1:2014 shall not be removable without the aid of a tool.		N/A
	For parts of Electric toys that contain batteries if the part fits wholly within the small parts cylinder as specified in 5.2 of ISO 8124-1:2014, the part shall not be removable without the aid of a tool.		N/A
	A force is applied to the part under consideration without jerks for 10 s in the most unfavourable direction. The force is as follows:		N/A
	– push force, 50 N;		N/A
	– pull force:		N/A
	• if the shape of the part is such that the fingertips cannot easily slip off, 50 N;		N/A
	• if the projection of the part that is gripped is less than 10 mm in the direction of removal, 30 N.		N/A
	The push force is applied by test probe 11 of IEC 61032. The pull force is applied by a suitable means, such as a suction cup, so that the test results are not affected. While the force is being applied, the test fingernail of Figure 7 of IEC 60335-1:2010, is inserted in any aperture or joint with a force of 10 N. The fingernail is then slid sideways with a force of 10 N but is not twisted or used as a lever.		N/A
	If the shape of the part is such that an axial pull is unlikely, the pull force is not applied but the test fingernail is inserted in any aperture or joint with a force of 10 N and is then pulled for 10 s by means of the loop with a force of 30 N in the direction of removal.		N/A
	If the part is likely to be twisted, the following torque is applied at the same time as the pull or push force:		N/A
	– 2 Nm, for major dimensions up to 50 mm;		N/A
	– 4 Nm, for major dimensions over 50 mm.		N/A

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Cl.	Requirement – Test	Result	Verdict
	This torque is also applied when the test fingernail is pulled by means of the loop. If the projection of the part which is gripped is less than 10 mm, the torque is reduced by 50 %.		N/A
	The part shall not become detached.		N/A
13.4.2	Other batteries		P
	Batteries shall not be removable without the aid of a tool unless the security of the battery compartment cover is adequate.		P
	An attempt is made to gain access to the battery compartment by manual means. It shall not be possible to open the cover unless at least two independent movements have to be applied simultaneously.		P
	The electric toy is placed on a horizontal steel surface. A cylindrical metallic mass of 1 kg, having a diameter of 80 mm, is dropped from a height of 100 mm so that its flat face falls onto the electric toy. The test is carried out once with the cylindrical metallic mass striking the electric toy in the most unfavourable place.		P
	The battery compartment shall not become open.		P
13.4.3	Electrolyte leakage		N/A
	Rechargeable batteries with liquid electrolyte shall not leak when the electric toy is placed in any position. The electrolyte shall not become accessible even if a tool has to be used to remove covers or similar parts.		N/A
13.4.4	Electric toys placed above a child		N/A
	Electric toys that are used with batteries where the intended fixed position of the battery compartment can be above a child shall have a battery compartment that prevents battery electrolyte leakage from the electric toy.		N/A
	All batteries are removed from the electric toy. The electric toy is placed in its normal orientation and the battery compartment is filled with the quantity of water specified in Table 2, the water being at a temperature of 21 °C ±5 °C.		N/A
	The electric toy's casing may be broken to gain access to the closed battery compartment in order to add water but any damage shall not affect the result of the test.		N/A

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Cl.	Requirement – Test	Result	Verdict
	After adding the water, the compartment is closed in accordance with the manufacturer's instructions taking care to avoid losing any water from the electric toy before the test is started. The electric toy is left in position for a period of 5 min.		N/A
	During the test, water shall not leak from the electric toy.		N/A
13.4.5	Parallel connection of batteries		N/A
	Batteries shall not be connected in parallel unless		N/A
	– the reverse insertion of batteries,		N/A
	– unbalanced discharging, or		N/A
	– unbalanced charging		N/A
	does not impair compliance with this standard.		N/A
	If screws or similar fasteners are used to secure a door or cover providing access to the battery compartment, the screw or similar fastener shall be captive to ensure that they remain with the door, cover or equipment.		N/A
	A force of 20 N is applied to the screw or similar fastener without jerks for a duration of 10 s in any direction.		N/A
	The screw or similar fastener shall not become separated from the door, cover or equipment.		N/A
13.5	Plug and sockets		
	Plugs and socket-outlets of Electric toys shall not be interchangeable with plugs and socket outlets listed in IEC TR 60083.		N/A
	This requirement is not applicable to plugs which are too large to be introduced into the mains socket outlets or that are too small so they can only be loosely inserted and do not stay firmly in place in the socket outlet aperture while in contact with the supply mains.		N/A
	Connectors such as jack types, USB types, RCA phono types with a diameter or diagonal measurement between 3,75 mm and 5,25 mm and length greater than 7 mm are considered to fail this requirement.		N/A
	Electric toys shall not use wires without connectors.		N/A
13.6	Charging batteries		N/A
	It shall be possible to charge secondary batteries inside the electric toy only if the following conditions are met		N/A

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Cl.	Requirement – Test	Result	Verdict
	– connection to, or replacement with primary batteries shall not be possible;		N/A
	– charging of other batteries or Electric toys from the electric toy shall not be possible;		N/A
	– connection of an incorrect polarity shall not be possible by constructions;		N/A
	– the battery charger shall comply with 15.4;		N/A
	– operation of the electric toy while charging shall not be possible unless the electric toy meets the requirements for Electric toys using a transformer or a power supply and the transformer or power supply complies with 15.3;		N/A
	– Electric toys for children under 3 years cannot operate while being charged.		N/A
	Mobile Electric toys shall not move during charging.		N/A
13.7	Series motors		N/A
	Electric toys shall not incorporate series motors having a power input exceeding 20 W.		N/A
13.8	Working voltage		N/A
	Internal parts of an electric toy having a working voltage exceeding 24 V shall not lead to any risk of harmful electric shock.		N/A
	In all conditions of test, the following values shall be met:		N/A
	– the working voltage between any two parts of the electric toy shall not exceed 5 kV when the electric toy is supplied at rated voltage;		N/A
	– the maximum current from a circuit with a generated voltage exceeding 24 V shall be less than 2 mA for DC and the peak value shall not exceed 0,7 mA for AC;		N/A
	– the capacitance of a circuit with a generated voltage exceeding 24 V and up to and including 450 V shall be less than 0,1 μF;		N/A
	– the discharge from circuits with a generated voltage exceeding 450 V and up to and including 5 kV shall not exceed 45 μC.		N/A
13.9	Electric toys connecting to other equipment		N/A
	Electric toys that can connect to class I equipment shall be safe when connected to that equipment in case of a fault in the equipment that the electric toy is connected to.		N/A

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Cl.	Requirement – Test	Result	Verdict
	Electric toys that can connect to class I equipment shall comply with one of the following conditions:		N/A
	a) the electric toy shall include an instruction to advise that the electric toy shall only be connected to equipment of Class II or class III (see 7.4); or		N/A
	b) conductive parts of the electric toy electrically connected to class I equipment shall not be accessible in the electric toy and the insulation between such parts and accessible parts shall have a thickness of at least 1 mm and an adequate electric strength.		N/A
	The electric toy is operated under normal operation according to 9.3. The electric toy is then disconnected from the supply and the insulation is immediately subjected to a voltage of 1 500 V having a frequency of 50 Hz or 60 Hz for 1 min, in accordance with IEC 61180.		N/A
	No breakdown shall occur during the test.		N/A
	For Electric toys that can connect to class I equipment complying with 13.9 b), the distances as stated in Clause 17 shall be fulfilled.		N/A
13.10	Speed limitation of ride-on Electric toys		N/A
	The maximum speed of ride-on Electric toys shall not exceed the limit specified in 4.23 of ISO 8124- 1:2014.		N/A
14	**PROTECTION OF CORDS AND WIRES**		N/A
14.1	Edges and moving parts		N/A
	Wireways are smooth and free from sharp edges.		N/A
	Cords and wires are protected so that they do not come into contact with burrs, cooling fins or similar edges that may cause damage to their insulation.		N/A
	Holes in metal through which cords and wires pass have smooth well-rounded surfaces or are provided with bushings.		N/A
	Cords and wires are effectively prevented from coming into contact with moving parts.		N/A
14.2	Bare wiring and heating elements are rigid and fixed so that during normal use clearances and creepage distances cannot be reduced below the values specified in Clause 18.		N/A
15	**COMPONENTS**		P

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Cl.	Requirement – Test	Result	Verdict
15.1.1	Components comply with the safety requirements specified in the relevant IEC standards as far as they reasonably apply.	See appended Table 15	P
15.1.2	Switches and automatic controls carrying a current exceeding 3 A during the tests of 9.3 and 9.4 comply with Annex C.		N/A
	Current (A)...:		-
	However, if they have been separately tested and found to comply with IEC 61058-1 or IEC 60730-1 respectively under the conditions occurring in the toy and for the number of cycles specified in Annex C, they may be used without further tests.		N/A
15.1.3	If components are marked with their operating characteristics, the conditions under which they are used in the toy are in accordance with these markings, unless otherwise specified.		P
	The testing of components that have to comply with other standards is, in general, carried out separately, according to the relevant standard.		P
	If the component is used within the limits of its marking, it is tested in accordance with the conditions occurring in the toy, the number of samples being that required by the relevant standard.		P
	When no IEC standard exists for the relevant component, when the component is not marked or is not used in accordance with its marking, it is tested under the conditions occurring in the toy. The number of samples is, in general, that required by a similar specification.......................................:		P
15.2	Prohibited components	No such components.	P
	Electric toys are not fitted with		P
	– thermal cut-outs that can be reset by a soldering operation;		P
	– mercury switches.		P
15.3	Transformers and power supplies		N/A
	Transformers for Electric toys comply with IEC 61558-2-7.		N/A
	Switch mode power supplies shall comply with IEC 61558-2-7 and IEC 61558-2-16.		N/A
	A battery charger that supplies an electric toy is considered to be also a power supply.		N/A

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Cl.	Requirement – Test	Result	Verdict
15.4	Battery chargers supplied with a toy comply with IEC 60335-2-29		N/A
	If they are battery chargers for use by children they comply with annex AA of that standard.		N/A
15.5	Batteries		N/A
	Primary batteries supplied with Electric toys shall comply with the relevant parts of the IEC 60086 series.		N/A
	Secondary batteries supplied with Electric toys shall comply with IEC 62133.		N/A
16	**SCREWS AND CONNECTIONS**		N/A
16.1	Fixings		N/A
	Fixings, the failure of which may impair compliance with these standard and electrical connections withstand the mechanical stresses occurring during play.		N/A
	Screws used for these purposes are not made of metal that is soft or liable to creep, such as zinc or aluminium.		N/A
	Screws used for electrical connections are screwed into metal.		N/A
	Screws and nuts are tested if they are used for electrical connections or are likely to be tightened by the user.		N/A
	The screws or nuts are tightened and loosened without jerking		N/A
	– 10 times, for screws in engagement with a thread of insulating material;		N/A
	– 5 times, for nuts and other screws.		N/A
	Screws in engagement with a thread of insulating material are completely removed and re-inserted each time.		N/A
	The test is carried out using a suitable screwdriver, spanner or key and by applying a torque as shown in Table 1.		N/A
	Column I is applicable for metal screws without heads if the screw does not protrude from the hole when tightened.		N/A
	Column II is applicable for other metal screws and for nuts and screws of insulating material.		N/A
	No damage impairing the further use of the fixings or electrical connections occur.		N/A
16.2	Connections		N/A

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Cl.	Requirement – Test	Result	Verdict
	Electrical connections carrying a current exceeding 0,5 A are constructed so that contact pressure is not transmitted through insulating material that is liable to shrink or to distort unless there is sufficient resiliency in the metallic parts to compensate for any possible shrinkage or distortion of the insulating material.		N/A
17	**CLEARANCES AND CREEPAGE DISTANCES**		N/A
	Clearances and creepage distances of functional insulation are not less than 0,5 mm except when the toy meets the requirements of Clause 9 with this distance short circuited.	Meets the requirements of Clause 9 with this distance short circuited.	N/A
	However, for functional insulation on printed circuit boards, except at their edges, this distance may be reduced to 0,2 mm provided that the degree of pollution in the microenvironment in which the insulation is located is unlikely to exceed pollution degree 2 during normal use of the toy.		N/A
	Internal parts of Electric toys that comply with subclause 13.8 and have a voltage exceeding 24 V have clearance and creepage distances for functional insulation equal to or greater than the values in Table 18 of IEC 60335-1:2010 for pollution degree 2 except when the toy meets Clause 9 with this distance short circuited.		N/A
	For guidance, the pollution degrees as defined in IEC 60335-1 are as follows:		N/A
	Degrees of pollution in the microenvironment:		N/A
	For the purpose of evaluating creepage distances, the following four degrees of pollution in the microenvironment are established		N/A
	– pollution degree 1: no pollution or only dry, non- conductive pollution occurs. The pollution has no influence;		N/A
	– pollution degree 2: only non-conductive pollution occurs, except that occasionally a temporary conductivity caused by condensation is to be expected;	Pollution degree 2	N/A
	– pollution degree 3: conductive pollution occurs or dry non-conductive pollution occurs that becomes conductive due to condensation that is to be expected;		N/A
	– pollution degree 4: the pollution generates persistent conductivity caused by conductive dust or by rain or snow.		N/A

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Cl.	Requirement – Test	Result	Verdict
	For Electric toys that can be connected to class I equipment both the creepage distance and clearance between accessible parts and conductive parts shall be at least 1,5 mm (see 13.9 b)).		N/A
18	**RESISTANCE TO HEAT AND FIRE**		P
18.1	Resistance to heat		N/A
	External parts of non-metallic material enclosing electric parts, and parts of insulating material supporting electric parts, are sufficiently resistant to heat if the toy has a working voltage exceeding 12 V and a current exceeding 3 A.		N/A
	The test is carried out at a temperature of 40 °C ±2 °C plus the maximum temperature rise determined during the tests of Clause 9 but it is at least 75 °C ±2 °C.		N/A
18.2	Resistance to fire		P
18.2.1	Parts of non-metallic material enclosing electric parts, and parts of insulating material supporting electric parts, are resistant to ignition and spread of fire.		P
	This requirement does not apply to decorative trims, knobs and other parts unlikely to be ignited or to propagate flames that originate from inside the toy.		N/A
	The tests are carried out on parts of non-metallic material that have been removed from the toy. When the glow-wire test is carried out, they are placed in the same orientation as they would be in normal use.		P
	These tests are not carried out on the insulation of cords and wires.		N/A
18.2.2	Non-metallic parts		P
	Parts of non-metallic material are subjected to the glow-wire test of IEC 60695-2-11, which is carried out at 550 °C.		P
	The glow-wire test is not carried out on parts of material classified at least HB40 according to IEC 60695-11-10, provided that the test sample was no thicker than the relevant part.		N/A
	Parts for which the glow-wire test cannot be carried out, such as those made of soft or foamy material, meet the requirements specified in ISO 9772 for category HBF material, the test sample being no thicker than the relevant part.		N/A
18.2.3	Insulating material		N/A

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Cl.	Requirement – Test	Result	Verdict
	Parts of insulating material supporting connections carrying a current exceeding 3A and having a working voltage exceeding 12 V, and parts of insulating material within a distance of 3 mm of such connections, are subjected to the glow-wire test of IEC 60695-2-11 at a temperature of 650 °C.		N/A
	However, the glow-wire test is not carried out on parts of material classified as having a glow-wire ignition temperature according to IEC 60695-2-13 of at least 675 °C, provided that the test sample was no thicker than the relevant part.		N/A
	Parts that withstand the glow-wire test of IEC 60695-2-11, but which, during the test, produce a flame that persists for longer than 2 s, are further tested as follows. Parts above the connection within the envelope of a vertical cylinder having a diameter of 20 mm and a height of 50 mm are subjected to the needle-flame test of Annex B. However, parts shielded by a barrier that meets the needle-flame test of Annex B are not tested.		N/A
	The needle-flame test is not carried out on parts of material classified as V-0 or V-1 according to IEC 60695-11-10, provided that the test sample was no thicker than the relevant part.		N/A
19	**RADIATION, TOXICITY AND SIMILAR HAZARDS**		N/A
19.1	Electric toys shall not emit harmful optical radiation or harmful electromagnetic radiation due to their operation in normal use.		N/A
19.2	Optical radiation		N/A
	Electric toys incorporating lasers and or light emitting diodes (LED) or UV emitting lamps shall comply with Annex E.		N/A
19.3	Other electromagnetic radiation		N/A
	Measurements methods for Electric toys with an integrated field source that may produce harmful electromagnetic radiation are given in Annex I.		N/A

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Cl.	Requirement – Test	Result	Verdict

ANNEX A	**EXPERIMENTAL SETS**		N/A
	The following modifications to this standard are applicable to all components of experimental sets supplied together or separately.		N/A
5	**GENERAL CONDITIONS FOR THE TESTS**		N/A
5.2	Preconditioning		N/A
	Not applicable		N/A
5.3	Assembly		N/A
	Addition: The tests are carried out with the experiments described in the instructions that result in the most unfavourable condition.		N/A

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Cl.	Requirement – Test	Result	Verdict
7	**MARKING AND INSTRUCTIONS**		N/A
7.3.4	Experimental sets		N/A
	The following warning shall be indicated on the packaging:		N/A
	WARNING: This toy is only intended for use by children over the age of X years (where X must be a minimum of 8)		N/A
	The substance of the following shall be indicated on the packaging:		N/A
	– an indication of the reasons for the age restriction;		N/A
	– that instructions for parents or care givers are included and shall be followed.		N/A
	The instructions for parents or care givers shall state the minimum age of the child for whom the set is intended.		N/A
	Detailed information shall be given in the instructions on how to set up and perform each experiment, indicating which phenomena are to be investigated. The instructions shall point out possible hazards and give technical information concerning the electronic components and electrical components, their behaviour and how to handle them properly. All hazards that can be expected during an experiment, such as those resulting from the short-circuiting of batteries or the wrong connection of capacitors, shall be described in detail.		N/A
	The instructions shall be written so that they are understandable by the age group for which the experimental set is intended.		N/A
	Instructions for children and for parents may be given separately. If the instructions are given in one leaflet, the section addressed to parents shall be given first.		N/A
	The instructions shall include a warning against manipulation of protective devices such as current- limiting devices. They shall describe the consequential dangers, such as overheating of cords, eruption of batteries and excessive heating.		N/A
8	**POWER INPUT**		P
	Not applicable.		P
9	**HEATING AND ABNORMAL OPERATION**		P
9.4	Not applicable.		P
9.6	Not applicable.		N/A

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Cl.	Requirement – Test	Result	Verdict
9.10	Addition: The temperature rise of surfaces, other than those of handles, knobs, buttons and similar parts, can exceed the limits if an appropriate warning is given in the instructions.		N/A
10	**ELECTRIC STRENGTH**		P
10.1	Electric strength at operating temperature		P
	Not applicable.		P
11	**ELECTRIC TOYS USED IN WATER, ELECTRIC TOYS USED WITH LIQUID AND ELECTRIC TOYS CLEANED WITH LIQUID**		N/A
	Not applicable.		N/A
12	**MECHANICAL STRENGTH**		P
	Not applicable.		P
13	**CONSTRUCTION**		P
13.1	Nominal supply voltage		
	Addition: The current does not exceed 5 A and the power input does not exceed 50 VA.		P
	Current (A)...:		-
	Power input (VA)................................:		-
	However these values may be exceeded during a period not exceeding 10 s.		P
	Period time (s)....................................:		-
14	**PROTECTION OF CORDS AND WIRES**		P
	Not applicable.		P
ANNEX B	**NEEDLE-FLAME TEST**	-	
	The needle-flame test is carried out in accordance with IEC 60695-11-5 with the following modifications.		P
7	**SEVERITIES**		P
	Replacement: The duration of application of the test flame is 30 s ±1 s.		P
9	**TEST PROCEDURE**		P
9.1	Position of test specimen		P
	Modification: The specimen is arranged so that the flame can be applied to a vertical or horizontal edge as shown in the examples of Figure 1.		P
9.2	Application of needle-flame		P
	Modification: The first paragraph does not apply.		P

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Cl.	Requirement – Test	Result	Verdict
	Addition: If possible, the flame is applied at least 10 mm from a corner.		P
9.3	Number of test specimens		P
	Replacement: The test is carried out on one specimen.		P
	If the specimen does not withstand the test, the test may be repeated on two additional specimens, both of which then withstand the test.		P
11	**EVALUATION OF TEST RESULTS**		P
	Addition: The duration of burning (t_b) does not exceed 30 s.		P
	However, for printed circuit boards, the duration of burning does not exceed 15 s.		P
ANNEX C	**AUTOMATIC CONTROLS AND SWITCHES**		N/A
C.1	Automatic controls		N/A
	Automatic controls that are tested with the toy comply with this standard and with subclauses 11.3.5 to 11.3.8 and Clause 17 of IEC 60730-1 as type 1 controls.		N/A
	The tests according to IEC 60730-1 are carried out under the conditions occurring in the toy.		N/A
	For the tests of Clause 17 of IEC 60730-1, the number of cycles of operation are		N/A
	– thermostats 3 000		N/A
	– self-resetting thermal cut-outs 300		N/A
	– non-self-resetting thermal cut-outs 10		N/A
C.2	Switches		N/A
	Mechanical switches that are tested with the electric toy shall comply with this standard and with the following clauses of IEC 61058-1-1, as modified below.		N/A
	Electronic switches that are tested with the electric toy shall comply with this standard and with the following clauses of IEC 61058-1-2, as modified below		N/A
	The tests of IEC 61058-1 and IEC 61058-1-2 are carried out under the conditions occurring in the toy.		N/A
	Before being tested, switches are operated 20 times without load.		N/A
12	**Construction**		N/A

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Cl.	Requirement – Test	Result	Verdict
	Switches are not required to be marked. However, a switch that can be tested separately from the appliance shall be marked with the manufacturer's name or trade mark and the type reference.		N/A
13	**Mechanism**		N/A
	NOTE The tests can be carried out on a separate sample.		N/A
15	**INSULATION RESISTANCE AND DIELECTRIC STRENGTH**		N/A
	Sub clause 15.1 is not applicable.		N/A
	Sub clause 15.2 is not applicable.		N/A
	Sub clause 15.3 is applicable for full disconnection and micro-disconnection.		N/A
17	**ENDURANCE**		N/A
	For 17.5.4, the number of cycles of actuation declared according to 7.4 is 3 000.		N/A
	Subclause 17.6.2 is not applicable.		N/A
	At the end of the tests, the temperature rise of the terminals shall not have increased by more than30 K above the temperature rise measured in Clause 9 of this standard. The tests may be carried out at the same time as the tests of Clause 9, provided the tests of Clause 9 are not affected.		N/A
	Temperature rise of the terminals (K)……………….:		N/A
20	**CLEARANCES, CREEPAGE DISTANCES, SOLID INSULATION AND COATINGS OF RIGID PRINTED BOARD ASSEMBLIES**		N/A
	This clause is applicable to clearances across full disconnection and micro-disconnection. It is also applicable to creepage distances for functional insulation, across full disconnection and micro- disconnection, as stated in Table 14.		N/A
Annex D	**ELECTRIC TOYS WITH PROTECTIVE ELECTRONIC CIRCUITS**		N/A
D.1	If during the tests of 9.9 an electronic circuit prevents the hazardous conditions listed in 9.10 or dangerous malfunction, it shall additionally comply with the following requirements. In this case, the electronic circuit is considered as a protective electronic circuit.		N/A

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Cl.	Requirement – Test	Result	Verdict
	If the protective electronic circuit includes only passive electronic components such as positive temperature co-efficient (PTC) resistors, negative temperature co-efficient (NTC) resistors or voltage dependent resistors (VDRs), the tests of Annex D are not applied.		N/A
D.2	Dangerous malfunction		N/A
D.2.1	The electric toy shall not malfunction in such a way as to cause an unintended operation that may impair safety or present a dangerous malfunction due to influence from electromagnetic phenomena (EMP).		N/A
	Electric toys using a transformer or a power supply where the electric toy incorporates a protective electronic circuit are additionally subjected to the tests of D.2.4 to D.2.8, using the supplied or the recommended transformer or power supply.		N/A
	The tests are carried out with the electric toy supplied at rated voltage and the electric toy operated in the following modes:		N/A
	– electronic off mode;		N/A
	– stand-by mode;		N/A
	– operating mode.		N/A
	The tests are carried out after the protective electronic circuit has operated during the fault conditions of 9.9.		N/A
	The tests are carried out with surge arresters disconnected, unless they incorporate spark gaps.		N/A
	Electric toys incorporating electronic controls complying with the IEC 60730 series are not exempt from the tests.		N/A
D.2.2	Electrostatic discharge		N/A
	The electric toy is subjected to electrostatic discharges in accordance with IEC 61000-4-2, test level 4 being applicable. Ten discharges having a positive polarity and ten discharges having a negative polarity are applied at each preselected point.		N/A
D.2.3	Radiated fields		N/A
	The electric toy is subjected to radiated fields in accordance with IEC 61000-4-3, test level 3 being applicable.		N/A
	The frequency ranges tested shall be 80 MHz to 1 000 MHz and 1,4 GHz to 2,0 GHz.		N/A

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Cl.	Requirement – Test	Result	Verdict
	The dwell time for each frequency shall be sufficient to observe a possible malfunction of the protective electronic circuit.		N/A
D.2.4	Transient bursts		N/A
	The electric toy is subjected to fast transient bursts in accordance with IEC 61000-4-4. Test level 3with a repetition rate of 5 kHz is applicable for signal and control lines. Test level 4 with a repetition rate of 5 kHz is applicable for the power supply lines. The bursts are applied for 2 min with a positive polarity and for 2 min with a negative polarity.		N/A
D.2.5	Voltage surges		N/A
	The power supply terminals of the electric toy are subjected to voltage surges in accordance with IEC 61000-4-5, five positive impulses and five negative impulses being applied at the selected points. Test level 4 is applicable for the line-to-line coupling mode, a generator having a source impedance of 2 Ω being used. Test level 4 is applicable for the line- to-earth coupling mode, a generator having a source impedance of 12 W being used.		N/A
	For Electric toys having surge arrestersincorporating spark gaps, the test is repeated at a level that is 95 % of the flashover voltage.		N/A
D.2.6	Injected current		N/A
	The electric toy is subjected to injected currents in accordance with IEC 61000-4-6, test level 3 being applicable. During the test, all frequencies between 0.15 MHz to 80 MHz are covered.		N/A
	The dwell time for each frequency shall be sufficient to observe a possible malfunction of the protective electronic circuit.		N/A
D.2.7	Voltage dips and interruptions		N/A
	The electric toy is subjected to the class 3 voltage dips and interruptions in accordance with IEC 61000-4-11. The values specified in Table 1 and Table 2 are applied to each test level, the dips and interruptions being applied at zero crossing of the supply voltage.		N/A
D.2.8	Mains signals		N/A
	The appliance is subjected to mains signals in accordance with IEC 61000-4-13:2002/AMD2:2015, Table 11 with test level class 2 using the frequency steps according to Table 10		N/A

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Cl.	Requirement – Test	Result	Verdict

ANNEX E	**SAFETY OF ELECTRIC TOYS INCORPORATING OPTICAL RADIATION SOURCES**		N/A
	The following modifications to this standard are applicable for Electric toys incorporating optical radiation sources, emitting in the wavelength range 200 nm to 3 000 nm.	Only indicator light	N/A
5	**GENERAL CONDITIONS FOR THE TESTS**		N/A
5.2	Addition: The tests of 19.E.2, 19.E.3 and 19.E.4 may be carried out on separate Electric toys.		N/A
5.3	Addition: The tests of 19.E.2, 19.E.3 and 19.E.4 are carried out before or after the preconditioning tests specified in 5.2, whichever is more unfavourable.		N/A
5.6	Addition: The tests of 19.E.2, 19.E.3 and 19.E.4 are carried out using the worst case emission taking the electric toy's function into account.		N/A
15	**Components**		N/A
15.2	Addition: Electric toys for children under the age of 3 years shall not incorporate lasers.		N/A
19	**Radiation, toxicity and similar hazards**		P
19.2	Addition:		P
	Electric toys shall not present a radiation hazard.		P
	Electric toys incorporating LEDs shall comply with 19.E.2.	Only indicator light	P
	Electric toys incorporating lasers shall comply with 19.E.3.		N/A
	Electric toys incorporating UV-emitting lamps shall comply with 19.E.4.		N/A
	All Electric toys incorporating optical radiation sources shall comply with 19.E.5.		N/A
19.E.2	Light-emitting diodes (LEDs)		P
	The emission from Electric toys incorporating LEDs shall not exceed the following limits:		P
	– 0,01 Wm-2 when assessed at 10 mm from the LED front for accessible emissions with wavelengths of < 315 nm;		N/A
	– 0,01 Wsr-1 or 0,25 Wm-2 when assessed at 200 mm, for accessible emissions with wavelengths of 315 nm ≤ λ < 400 nm;		N/A

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Cl.	Requirement – Test	Result	Verdict
	– 0,04 Wsr-1 or the AEL specified in Tables E.2 or E.3 assessed at 200 mm for accessible emissions with wavelengths of 400 nm ≤ λ< 780 nm;		N/A
	– 0,64 Wsr-1 or 16 Wm-2 when assessed at 200 mm for accessible emissions with wavelengths of 780 mm ≤ λ < 1 000 nm;		N/A
	– 0,32 Wsr-1 or 8 Wm-2 when assessed at 200 mm for accessible emissions with wavelengths of1 000 nm ≤ λ< 3 000 nm.		N/A
19.E.2.1	Measurement of emission from Electric toys		N/A
19.E.2.2.1	UVB and UVC AEL		N/A
19.E.2.2.2	UVA AEL		N/A
19.E.2.3	Visible light AEL		N/A
19.E.2.4	Infrared AEL		N/A
19.E.2.5	Groups of LEDs	Only indicator light	P
19.E.3	Lasers		N/A
	Lasers in Electric toys shall not exceed the AEL for class 1 laser products when measured in accordance with Clause 4 and 5 of IEC 60825-1:2014 using measurement conditions in IEC TR 60825-13 where appropriate.		N/A
19.E.4	UV-emitting lamps		N/A
	Emission from Electric toys incorporating UV- emitting lamps shall not exceed the limits for wavelength λand assessment distance δas follows. λ < 315 nm δ = 10 mm Children 3 years or younger 0,001 Wm λ < 315 nm δ = 10 mm Children older than 3 years 0,01 Wm-2 315 nm ≤ λ ≤ 400 nm δ = 200 mm Children 3 years or younger 0,02 Wm-2 315 nm ≤ λ ≤ 400 nm δ = 200 mm Children older than 3 years 0,2 Wm-2		N/A
19.E.4.1	Measurement of emission from Electric toys		N/A
19.E.5	Modulated accessible emission		N/A
	The packaging or instructions for Electric toys with modulated output from visible optical radiation sources with a frequency of modulation between 4 Hz and 60 Hz shall include the following warning that shall be visible at the point of purchase:		N/A
	WARNING: This toy produces flashes that may trigger epilepsy in sensitised individuals.		N/A
ANNEX I	**ELECTRIC TOYS GENERATING ELECTROMAGNETIC FIELDS (EMF)**		N/A

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Cl.	Requirement – Test	Result	Verdict
	For Electric toys with an integrated field source generating EMF the measurement methods in IEC 62233: 2005 are applicable with the following modifications.		N/A
A.1	Addition: Measurements are not necessary for Electric toys – without a motor or inductor; – which only include passive electronic components; or; – having an input current not exceeding 3 A.		N/A
	The current is checked by measurement during the tests of IEC 62115:2016, 9.3 unless the construction of the electric toy is such that the current cannot exceed 3 A.		N/A
A.2.3	Measuring distance and sensor location		N/A
	Modification: Add the following to Table A.1 before the row for "Tumble dryers":		N/A

[illegible]	[illegible]	All surfaces	Continuously
[illegible]	[illegible]	All surfaces	Continuously
[illegible]	[illegible]	All surfaces	Continuously

ANNEX J	**SAFETY OF REMOTE-CONTROLS FOR ELECTRIC RIDE-ON ELECTRIC TOYS**		N/A
7	**Marking and instructions**		N/A
7.J.1	Remote control ride-on Electric toys		N/A
	Remote control for electric ride-on Electric toys, both as a separate add-on product, or, when integrated into the electric ride-on toy, shall comply with the requirements given in 7.J.1.1, 7.J.1.2, 7.J.1.3 and 7.J.1.4.		N/A
7.J.1.1	Documentation		N/A
	The system documentation accompanying the electric ride-on toy shall include the following documents:		N/A
7.J.1.2	Instructions for use		N/A
	The markings required by 7.5 of the main body of the standard shall be repeated in the instructions for use. The instructions for use shall also include warnings and a clear instruction as follows:		N/A

	WARNING: The remote control is not a toy. It is for the adult use only, and should not be used by a child. A close adult supervision is always required. Reception range may change significantly with weather, battery, and other environmental conditions.		N/A

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Cl.	Requirement – Test	Result	Verdict
	The instructions for use shall also include a full list of controlled modes, and detailed operating instructions for any such operation mode.		N/A
	If the product has an uncontrolled mode, then the instructions for use shall also include an explanation detailing all possible methods of switching the system from controlled mode to uncontrolled mode.		N/A
	Methods of switching shall comply with the requirement of 13.J.1.2.		N/A
7.J.1.3	Installation instructions, compatible electric ride-on Electric toys listing, electrical specification		N/A
	In the event that the remote control system is intended to be sold as a separate add-on product, the documentation shall include:		N/A
	a) listing of electric ride-on toy models, which the remote control is compatible with;		N/A
	b) installation instructions for the receiver in the compatible electric ride-on toy;		N/A
	c) specification of maximum permitted voltage and current load.		N/A
7.J.1.4	Manufacturer declaration		N/A
	Manufacturer declaration shall include a clarification how mutual operation is prevented. The declaration shall be in line with the requirements of 9.J.1.2.		N/A
9	**Heating and abnormal operation**		N/A
9.J.1	Remote control ride-on Electric toys		N/A
	Remote control for electric ride-on Electric toys, both as a separate add-on product, or, when integrated into the electric ride-on toy, shall also comply with the requirements given in 9.J.1.1 and 9.J.1.2.		N/A
9.J.1.1	Automatic stop upon wireless communication cut- off		N/A
	In the event that wireless communication between the transmitter and the receiver is cut-off, the electric ride-on toy shall stop automatically. Furthermore, in any event of a system failure or malfunction that prevents the adult's ability to control the electric ride-on toy as required in 13.J.1.1, the electric ride-on toy will stop by default.		N/A
	The braking performance shall comply with 4.21(b) and 5.16.2 of ISO 8124-1:2014.		N/A

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Cl.	Requirement – Test	Result	Verdict
	The requirements specified in this clause are applicable only in controlled mode. They do not apply in uncontrolled mode, if such a mode exists.		N/A
	Compliance is checked by the following test:		N/A
	a) Place the electric ride-on toy on the test bench. If necessary, elevate the electric rideon toy wheels above the bench surface, to enable simulation of driving and stopping, without moving from thespot.		N/A
	b) While simulating the electric ride-on toy driving, cause intentional wireless communication cut-off as follows: disconnect the transmitter from its power source and discharge its power stabilizing capacitors if any.		N/A
	c) Verify that the electric ride-on toy stops automatically.		N/A
	d) Verify the braking performance meets the requirements defined in 4.21(b) and 5.16.2 of ISO 8124-1:2014.		N/A
	e) Repeat the process for each of the controlled modes, as listed in the instructions for use provided as required in 7.J.1.2.		N/A
	In the event that the remote control system is intended to be sold as a separate add-on product, the test shall be performed with the maximum permitted voltage and current load, as defined in the specification provided as required in 7.J.1.3. In this case, the current load shall be applied in the most unfavourable way of operation given in the operating instructions.		N/A
9.J.1.2	Prevention of mutual operation		N/A
	The transmitter of one electric ride-on toy shall not influence the control of another electric ride-on toy. The amount of transmitter variants of the remote control systems of the same type shall be at least 4,096 and the utilization of these variants will be cyclic during production.		N/A
	However, in case that a malfunction in one remote control is causing interference to another electric ride-on toy, then the other one could stop due to communication cut-off, as specified in 9.J.1.1.		N/A
13	**Construction**		N/A
13.J.1	Remote-control ride-on Electric toys		N/A
	Remote control for electric ride-on Electric toys, both as a separate add-on product, or, when integrated into the electric ride-on toy, shall comply with the following requirements:		N/A

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Cl.	Requirement – Test	Result	Verdict

13.J.1.1	Ability to control the electric ride-on toy by means of the remote control		N/A
	The adult shall be able to control the electric ride- on toy by means of operating control buttons in the transmitter. As a minimum requirement, the control function shall provide the ability to stop the electric ride-on toy.		N/A
	The braking performance shall comply with 4.21(b) and 5.16.2 of ISO 8124-1:2014.		N/A
	The requirements specified in 13.J.1.1 are applicable only in controlled mode. They do not apply in un-controlled mode, if such a mode exists.		N/A
	Compliance is checked by the following test:		N/A
	a) Place the electric ride-on toy on the test bench. If necessary, elevate the electric rideon toy wheels above the bench surface, to enable simulation of driving and stopping, without moving from thespot.		N/A
	b) While simulating the electric ride-on toy driving, activate the stop function of the remote control, as specified in the instructions for use provided as required in 7.J.1.2.		N/A
	c) Verify that the electric ride-on toy stops as required.		N/A
	d) Verify the braking performance meets the requirements defined in 4.21(b) and 5.16.2 of ISO 8124-1:2014.		N/A
	e) Repeat the process for each of the controlled modes, as listed in the instructions for use provided as required in 7.J.1.2.		N/A
13.J.1.2	Restriction of switching from controlled mode to uncontrolled mode		N/A
	If the product has the option of switching the system from controlled mode to uncontrolled mode, then such switching shall only be intended to be adjusted by an adult by means of a tool.		N/A
13.J.1.3	Avoidance of any control function in uncontrolled mode		N/A
	In uncontrolled mode, the remote control shall not have any effect on the electric ride-on toy.		N/A

8	**TABLE: Input data under normal operation**				**N/A**
	Rated voltage U (V)	Rated input (W) or (VA)	Measured input (W) or (VA)	Deviation	Normal operation / Remarks

Single Voltage (V)	Lower Voltage Limit (V)	Upper Voltage Limit (V)	Mean Value of Range				
--	--	--	--	--	--	--	--
--	--	--	--	--	--	--	--
Supplementary information:							

9.3	TABLE: Heating Test (Normal operation)		P
	Test voltage (V) ....:	3.7Vdc	
	Ambient (°C) ....:	24.8 °C	
	Operating time ....:	2h10mins	
	Input Watts (W)..:		
	Input Volt-Amperes (VA) ....:		

Thermocouple Locations	Max.Temperature rise (K)	Max. temperature limit, (K)
PCB near internal wire	8.9	85
Internal wire	4.9	35
Battery surface	5.2	Ref.
Plastic enclosure inside near PCB	5.1	Ref.
Plastic enclosure outside near PCB	4.3	35
Supplementary information: The maximum operating temperature is +45°C.		

9.4	TABLE: Heating Test (Normal operation with insulation short-circuited)		N/A
	Test voltage (V)..:	--	
	Ambient (°C)..:	--	
	Operating time ....:	--	
	Input Watts (W) ....:	--	
	Input Volt-Amperes (VA)..................................:	--	

Thermocouple Locations	Max.Temperature rise (K)	Max. temperature limit, (K)
--	--	--
--	--	--
Supplementary information: see table 9.9		

9.5	TABLE: Heating Test (Abnormal operation with temperature controls made inoperable)		N/A
	Test voltage (V) ....:	--	
	Ambient (°C) ....:	--	
	Operating time ....:	--	
	Input Watts (W)..:	--	
	Input Volt-Amperes (VA) ....:	--	

Thermocouple Locations	Max.Temperature rise (K)	Max. temperature limit, (K)
--	--	--
--	--	--

Supplementary information:
The test of 9.3 is repeated with any device limiting the temperature during the tests of 9.3 being disabled. If the electric toy has more than one control, they are disabled in turn.

9.6	TABLE: Heating Test (with accessible moving parts locked)		N/A
	Test voltage (V) ....:	--	
	Ambient (°C) ....:	--	
	Operating time ....:	--	
	Input Watts (W)..:	--	
	Input Volt-Amperes (VA) ....:	--	

Thermocouple Locations	Max.Temperature rise (K)	Max. temperature limit, (K)
--	--	--
--	--	--

Supplementary information:

1) The test of 9.3 is repeated with accessible moving parts locked.
2) If the electric toy incorporates more than one motor, the test is carried out by locking moving parts driven by each motor in turn.

9.8	TABLE: Heating Test (Abnormal supply to Electric toys via a USB connection)		N/A
	Test voltage (V) ....:	--	
	Ambient (°C) ....:	--	
	Operating time ....:	--	
	Input Watts (W)..:	--	
	Input Volt-Amperes (VA) ....:	--	

Thermocouple Locations	Max.Temperature rise (K)	Max. temperature limit, (K)
--	--	--
--	--	--

Supplementary information: For Electric toys supplied with power from a USB connection, the test of 9.3 is repeated with the toy being supplied with a voltage of 42 V.

9.9	TABLE: Fault Condition Tests				P
	Ambient temperature (C)..............................:				
Component	Fault Condition	Test Voltage (V)	Test Duration	Fuse-link Current (A)	Comment/Result
Battery	SC	6V7AH	10mins	--	Unit cannot be worked as normally. After test, no damaged, no hazard.
Supplementary Information: SC: Short circuits					

12	TABLE: Impact Resistance		P
Impacts per surface	**Surface tested**	**Impact energy**	**Comments**
Three blows	Top surface	0.5J	No damaged, no cracks
Three blows	Side surface	0.5J	No damaged, no cracks
Three blows	Bottom surface	0.5J	No damaged, no cracks
Supplementary information:			

16.1	TABLE: Threaded Part Torque Test		N/A
Threaded part identification	Diameter of thread (mm)	Column number (I or II)	Applied torque (Nm)
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Supplementary information:			

17	TABLE: Clearance And Creepage Distance Measurements					N/A
clearance cl and creepage distance dcr at/of:	Up (V)	U r.m.s. (V)	Required cl (mm)	cl (mm)	required dcr (mm)	dcr (mm)
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Supplementary information:						

18.1	TABLE: Ball Pressure Test of Thermoplastics		N/A
Allowed impression diameter (mm) ....:			
Object/ Part No./ Material	Manufacturer/ trademark	Test temperature (C)	Impression diameter (mm)
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Supplementary information: The test is carried out at a temperature of 40 °C ±2 °C plus the maximum temperature rise determined during the tests of Clause 9 but it shall be at least 75 °C ±2 °C.			

18.2.2	TABLE: Glow Wire Test (GWT)					P
Test Conditions....................	GWT according to IEC 60695-2 11 (for non-metallic parts)					
Testtemperature (°C)..........	550 C					
Object/ Part No./ Material	Manufacturer/ trademark	te,	ti	Specified Layer under Test Specimen ignited, Yes/No	Other remarks	
Enclosure	--	6.0s	2.0s	No	--	
Supplementary information: Parts of non-metallic material are subjected to the glow-wire test of IEC 60695-2- 11, which is carried out at 550 °C. ti: the duration, ti (to the nearest 0,5 s), from the beginning of tip application up to the time at which the test specimen or the specified layer placed below it ignites; te: the duration, te, (to the nearest 0,5 s) from the beginning of tip application up to the time when all flames extinguish, during or after the period ofapplication;						

18.2.3	TABLE: Glow Wire Test (GWT) / Glow Wire Ignition Temperature (GWIT)					N/A
Test Conditions....................	GWT according to IEC 60695-2 11 (for insulation material)					
Testtemperature (°C)..........	650 C					
Object/ Part No./ Material	Manufacturer/ trademark	te,	ti	Specified Layer under Test Specimen ignited, Yes/No	Other remarks	
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Object/ Part No./ Material	Manufacturer/ trademark	GWIT		--	Other remarks	
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Supplementary information:

Parts of insulating material supporting connections carrying a current exceeding 3A and having a working voltage exceeding 12 V, and parts of insulating material within a distance of 3 mm of such connections, are subjected to the glow-wire test of IEC 60695-2- 11 at a temperature of 650 °C. However, the glow-wire test is not carried out on parts of material classified as having a glow-wire ignition temperature according to IEC 60695-2- 13 of at least 675 °C, provided that the test sample was no thicker than the relevant part.

ti: the duration, ti (to the nearest 0,5 s), from the beginning of tip application up to the time at which the test specimen or the specified layer placed below it ignites;

te: the duration, te, (to the nearest 0,5 s) from the beginning of tip application up to the time when all flames extinguish, during or after the period ofapplication;

18.2.3	TABLE: Needle- flame test (NFT)				N/A
Object/ Part No./ Material	Manufacturer/ trademark	Duration of application of test flame (ta); (s)	Ignition of specified layer Yes/No	Duration of burning (tb) (s)	Verdict
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Supplementary information:

NFT not relevant for Parts of material classified as V-0 or V-1

ANNEX A:

Photo-documentation

Photo 1

Photo 2

Photo 3

Photo 4

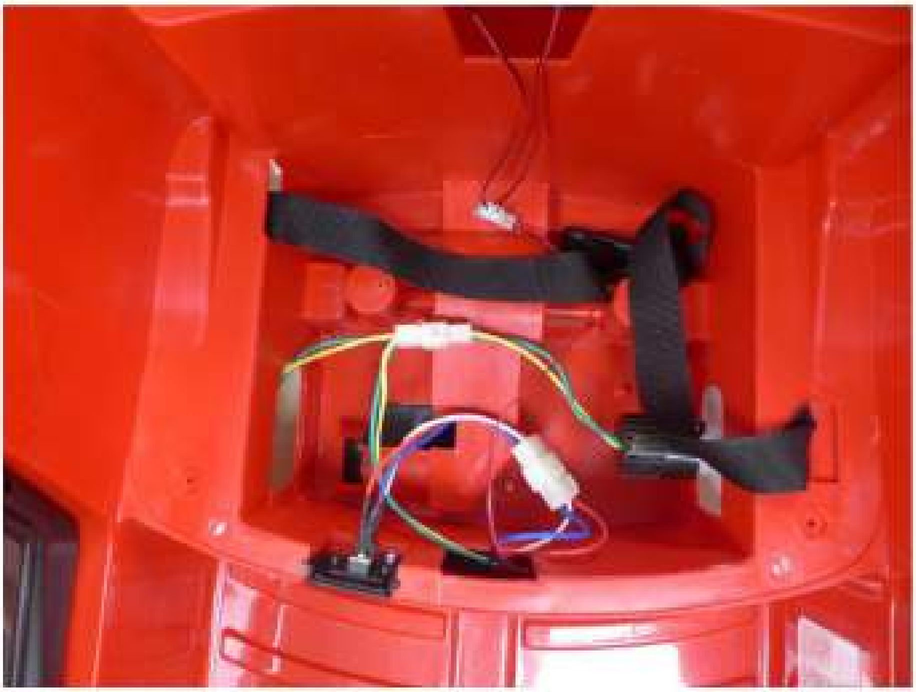

Photo 5

Photo 6

Photo 7

Photo 8

Photo 9

Photo 10

Photo 11

Photo 12

Photo 13

Photo 14

Photo 15

Photo 16

Photo 17

Photo 18

Photo 19

Photo 20

Photo 21

Photo 22

Photo 23

Photo 24

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Photo 35

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